

Zhenyu He

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EDUCATION

Graduate Study at UC, Berkeley

Earth and Planetary Science Department. PI: David Romps; Secondary PI: Inez Fung
High Performance Fluid Dynamics Simulations, Climate Physics

Berkeley, USA, 08/2022 – Present

Bachelor at Peking University

School of Physics, Department of Atmospheric and Oceanic Sciences, Top 3%
Double Degree in Economics, Top 1%

Beijing, China, 09/2018 – 06/2022

PUBLICATIONS

Agents' Last Exam (ALE) (co-author, Data Contributors group). A 1,000+-task agent benchmark across 55 specialized domains, from UC Berkeley RDI (Dawn Song lab). Built and evaluated agent tasks in the natural-science domain. Submission under review, NeurIPS 2026 Evaluations and Datasets (E&D) Track. arXiv:2606.05405.

MANUSCRIPTS UNDER REVIEW

"A cloud-resolving simulation of the Hiroshima firestorm plume." Zhenyu He (first author); corresponding author David Romps. Under review at JGR-Atmospheres, 2026.

"A lower bound on the altitude of the Hiroshima firestorm plume from ray optics." Zhenyu He (third author); corresponding author David Romps. Under review at Geophysical Research Letters (GRL), 2026.

RESEARCH EXPERIENCE

High-Resolution 3D LES of the Hiroshima Firestorm Plume

Graduate Student Researcher, Supervised by Prof. David Romps, UC Berkeley

09/2024 – 09/2025

- Developing the first high-resolution 3D LES (with water) of the Hiroshima firestorm plume in DAM, combined with photogrammetric evidence to constrain plume-top height near the tropopause.
- Designed a tractable sensitivity plan (best-guess + targeted perturbations) across soundings, fire radius R , and heating rate H , acknowledging compute limits ($\approx 46k$ core-hours per run).
- Implemented and debugged DAM code extensions to output qv_source and Q_latent diagnostics and to control fallout/evaporation (e.g., $precip_fall_factor$, $evap_factor$) for process attribution.
- Built a workflow to ingest spatially varying heat-flux maps (digitized from published figures) into DAM via new forcing logic ($heatflux_do + read_conforming$), and optimized I/O cadence for runtime efficiency.
- Compiled and cross-validated historical constraints on fire geometry and meteorology (e.g., reported burn radius ranges; wind profiles from declassified mission reports) to ground the modeling configuration.

Global Monthly Diagnostic Pipeline for Plume Dynamics

Graduate Student Researcher, Supervised by Prof. David Romps, UC Berkeley

09/2022 – 09/2024

- Built a global monthly diagnostic pipeline mapping LZRH, LNB, and LNB–LZRH under fixed plume radius and heat source, and evaluated where plumes can exceed radiative-heating constraints.
- Performed sensitivity sweeps over heat source H and plume radius R , identifying how ascent limits and $w=0/LNB$ respond to scaling assumptions.
- Diagnosed numerical stability: demonstrated strong dependence of LNB on integration step size and selected a converged step (e.g., 1 m) for robust results.
- Quantified sensitivity to initialization height (e.g., 100 m vs 500 m vs 1000 m) and explained changes via buoyancy/entrainment scaling.
- Reproduced and validated published benchmark behavior by correcting pressure/temperature profile inconsistencies.
- Compared plume outcomes across different historical profiles with different stability, documenting structural differences and limits imposed by sounding top height.

Whether Nuclear Winter Becomes Reality When Nuclear Exchanges Happen Around the Globe

Independent Research, Supervised by Prof. David Romps, UC Berkeley

12/2022 – 03/2024

- Developed 1-D model of vertical motion of air parcel with different initial conditions and environmental profiles.
- Compared different indices for the tropopause and set the level of net zero radiative heating as the threshold of soot gaining much longer lifetime.

Sentinel-5P/TROPOMI Methane Anomaly Mapping over ConUS

Graduate Student Researcher, Supervised by Prof. Inez Fung, UC Berkeley

12/2023 – 06/2024

- Built a Sentinel-5P/TROPOMI L2 CH₄ processing pipeline (netCDF → QA filtering → spatial remapping) to produce monthly/annual methane anomaly maps over ConUS.
- Implemented background-corrected anomaly definition using latitude-band daily medians; evaluated sensitivity to domain and time-window choices.
- Accelerated large-scale coordinate matching via KD-tree spatial indexing and gridded aggregation for consistent temporal averaging.
- Designed interpretable regional experiments (cattle-rich vs rice-rich vs oil-well-rich) to identify distinct temporal signatures before scaling globally.
- Benchmarked satellite-derived spatial patterns against EPA gridded methane emissions products to diagnose discrepancies and attribution limits.

The Development of a Joint Aerosol Retrieval Algorithm for MODIS and CALIPSO Sensors

Independent Research, Supervised by Prof. Jing Li, Peking University

10/2021 – 06/2022

- Replaced lookup tables of MODIS with retrieved aerosol extinction coefficient profile by CALIPSO.
- Added retrieved AOD by MODIS as the boundary condition for retrieving aerosol extinction coefficient by CALIPSO.
- Iterated the above steps until the results of retrieved AOD by MODIS and the aerosol extinction coefficient profile by CALIPSO are stable.

Markov Chain Monte–Carlo (MCMC) to Do Retrievals

Independent Research, Supervised by Prof. Yuk Yung, Caltech and Prof. King-Fai Li, UCR

03/2021 – 03/2022

- Used MCMC to solve the weakness of Rodgers' method to do retrievals.
- Examined the MCMC method for the Earth Science applications.
- Tried to retrieve the land and ocean distribution of Earth from DSCOVR with single-point light curve.
- Applied different methods to find best regularization parameters to solve ill-conditioned problem in retrievals.

Dependence of Walker Circulation on the Width of Ocean Basin

Independent Research, Supervised by Prof. Yongyun Hu and Prof. Ji Nie, Peking University

05/2020 – 06/2021

- Explored the dependences of Walker circulation on the width of ocean basin with idealized aquaplanet simulations.
- Developed diagnostic methods to reveal the controlling mechanisms of the above dependence.
- Examined and understood Walker circulation in paleo-climate simulations with realistic geography.

The Response of Walker Circulation and Hadley Circulation Under Global Warming

Independent Research, Supervised by Prof. Ji Nie, Peking University

10/2019 – 05/2020

- Found the change of the intensity of Walker Cell and Hadley Cell with different levels of CO₂.
- Calculated the different contributions of Walker Cell and Hadley Cell to the total change of mass flux at tropical areas.

INVITED TALK

Princeton School on Science and Global Security

Invited Speaker, Princeton University

10/2024

- Selected as one of the next-generation scientists to engage with technical perspectives on nuclear weapon threats and global security.
- Awarded full travel and accommodation funding by the Program on Science and Global Security (SGS).
- Presented research: "Constraining the Dynamics of Extreme Convection in the Case of Hiroshima" — utilizing atmospheric fluid dynamics (MTT model & LES) to re-evaluate the "Nuclear Winter" hypothesis and stratospheric soot injection mechanisms.

AWARDS AND HONORS

Princeton SGS Scholarship	10/2024
H2H8 Scholarship	11/2025
Petroleum Scholarship (3%)	10/2019
Second Prize of Outstanding Volunteers of School of Physics, Peking University (4%)	11/2020
Huatai Securities Technology Scholarship (3%)	10/2021

EXTRACURRICULAR ACTIVITIES

Homecoming High School Practice Group, school-level first prize

Co-leader

11/2018 – 02/2019

- Recruited volunteers from different majors, collected information on major features, learning experience, career development etc.
- Conducted lectures, launched a series of activities about introducing Peking University in high schools.
- Won a school-level first prize in the Homecoming volunteers' activities.

Volunteer Teaching in Majiang Middle School, Qishan, Shaanxi

Leader

06/2020 – 09/2020

- Formed a team of Chinese college student volunteers to the remote and poor areas and conducted "Helping the Will and Cultivating the Wisdom" activities.
- Gave lectures to students of Majiang Middle School and supervised self-study courses.
- Received positive feedbacks from 93% of participating students.

Volunteer Club of PKU

Core Member

09/2018 – 06/2022

- Led volunteer teams to the Angle House, took care of handicapped children every week, taught them to paint and sing songs, etc.

Student Union, School of Physics, PKU

Minister

09/2019 – 09/2020

- Maintained good relationship with sustaining enterprise partners such as Valeon, DH International, achieved sponsorship funds about \$11,000 per year, did the annual financial report within the department.
- Organized cross-department activities such as balls, long distance carbon-reduction running, etc. Confirmed the number of participants, funds and plans with different departments before activities, and prepared the contingency plan for emergency situations.
- Organized the design of department shirts, planned for the cultural festival, exhibited the department culture to students in other departments.

Class 4 of Grade 2018, School of Physics, PKU

Monitor

09/2018 – 06/2022

Baseball Team, PKU

Third Baseman

09/2018 – 09/2021

ONE-AGI Department

Department Head

2025 – Present

- Lead a department focused on empowering individuals in the AI era: leveraging cutting-edge AI tools to achieve 10x efficiency gains and expand personal boundaries into previously unattainable domains.
- Facilitate discussions on critical issues surrounding AGI, including safety, societal impact, legal frameworks, individual freedom, and the path toward harmonious human-AGI coexistence.
- Organize workshops and seminars on self-improvement strategies, AI-augmented productivity, and responsible AI adoption.